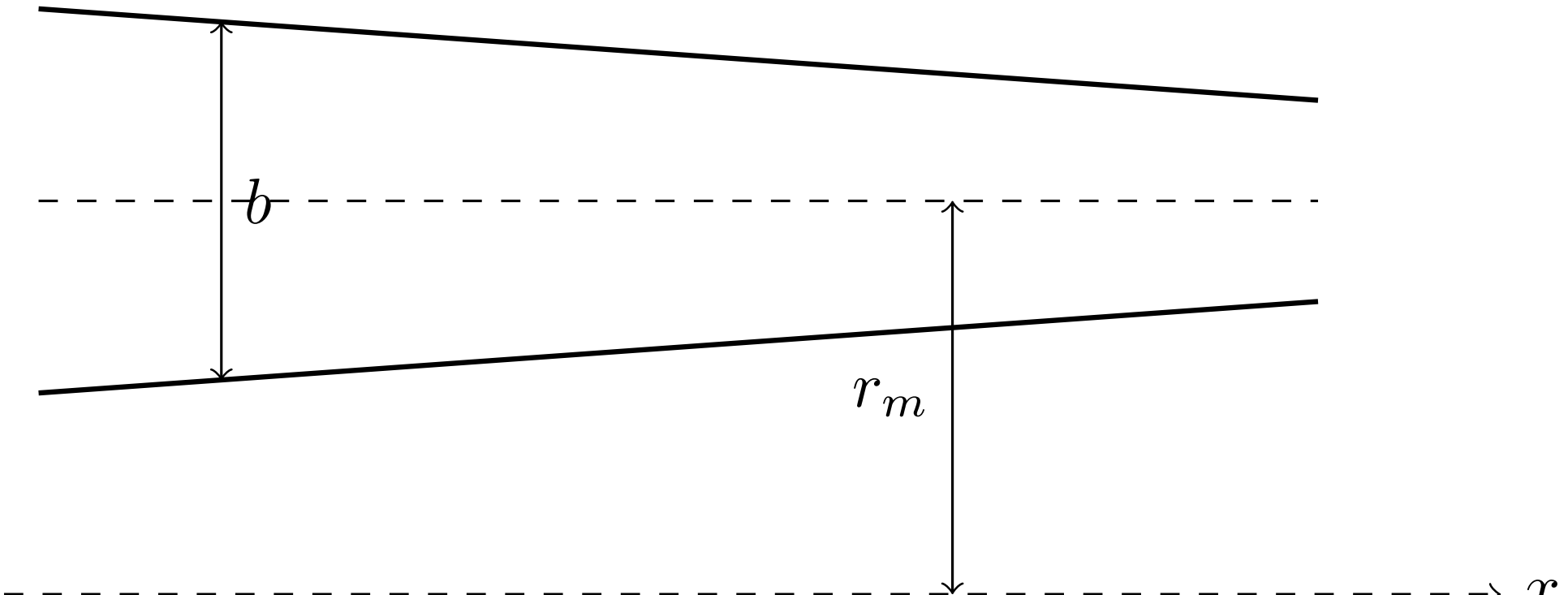$$A = \pi(r_t^2 - r_h^2) = 2\pi r_m b$$